

Trustworthy Machine Learning Assignment 1

Team number - LV

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Stolen Model Detection

For this assignment, we are given access to the target model weights, along with their information and white-box access to 360 suspect model weights and we have to find out which models are directly stolen, or tweaked from the target model. This can be done through a stealing confidence score. Higher Score implies Suspect models derived from target models in some capacity.

The Process

We used these methods consecutively to figure out the suspected models.

1) Weight Similarity

For the first method, we went the basic route which is just to consider the weights of the models. To achieve this- we used these two parameters- Cosine similarity and L2 similarity. The idea for this method is that copied models usually have similar parameter values. Cosine similarity calculates directional difference while the L2-based similarity calculates distance between weight vectors.

2) BatchNorm Similarity

Batch Normalization Statistics consists of Mean and Variance. They are more of internal spies of the training process. If batch normalization statistics of two models tells us that the models were trained from the same source or copied from one another.

3) Behavioural Analysis

Now the focus is on model behavior rather than parameters. For this, CIFAR-100 test set is used as a testing dataset. These same images in the CIFRA-100 were tested through both the target model and suspect model. The intuition is that if two models give similar output for the testing images, they might've had to use similar knowledge to train.

4) Multi-Layer Feature Similarity

Since the model has several layers, we look at the results of different layers. Even after some possible changes, stolen models usually generate feature representations which are similar to the target model.

The results of the penultimate layer and the target are compared and calculated using cosine similarity. It is also an internal level process since it calculates similarity at different layers inside the models.

5) Rank Fusion Computation

All of the four methods gives a raw score, but their scales differ. Instead of normally averaging them, we used rank fusion for each method, all 360 suspect models were ranked from most likely stolen to least likely stolen. Combination of those rankings is then turned into a single final score. The final score for each suspect model is derived from this combined ranking.

Results

Our first submission achieved a score of 0.1667 (TPR @ 5% FPR). To improve this score, we first added more methods to calculate the difference scores and thereafter, to combine all the methods used, we used the the rank fusion approach.

Final Score: **0.629630**

Conclusion

Thus, we built a flow that determines stolen models by combining four different methods. The combined result of these methods were able to detect potentially stolen models.

- Weight similarity
- BatchNorm Similarity
- Behavioral Analysis
- Multi Layer Feature Similarity
- Rank Fusion Computation

This assignment shows that Model theft implications are serious. A model can be stolen easily by a third party and deployed without bearing any of the initial training cost and resources which is unfair. This can also leads to Privacy leaks from the biases,weights and other parameters of the training data.